

NeuroLens

An Elastic Cloud-Native Neuromarketing SaaS for Video Engagement Prediction

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1. Motivation & Business Case

1.1 The Power and Inaccessibility of Neuromarketing

Neuromarketing, the application of cognitive neuroscience to decode how the brain responds to advertising, is one of the most direct methods of commercial evaluation. When viewers watch an ad inside an fMRI scanner, second-by-second changes in cortical blood oxygenation capture attentional and emotional responses that traditional self-report methods cannot: they suffer from post-hoc rationalisation, social desirability bias, and memory decay. Venkatraman et al. (2015) demonstrated that fMRI measures significantly outperformed conventional survey and biometric instruments in predicting real-world advertising sales elasticity. Yet physical fMRI requires multi-million-dollar 3-Tesla scanners and clinical staff, costing \$10,000–\$15,000 per commercial, placing it out of reach for most studios and agencies.

1.2 The Modern Catalyst: Faster Production, Harder Evaluation

Video advertising is now dominated by fast-scrolling mobile feeds such as TikTok, Instagram Reels, and YouTube Shorts, where viewers can dismiss any video with a swipe. The opening hook matters, but the challenge extends beyond the first few seconds: even a well-hooked viewer can be lost at a jarring transition or a mid-video lull.

Simultaneously, generative AI tools such as Runway, Sora, and CapCut allow creators and marketing teams to produce numerous creative variants rapidly. This creates a Production–Evaluation Asymmetry: generating alternatives is fast and inexpensive, but understanding where and why a creative might lose its audience (at any point in the timeline) before publication remains difficult.

1.3 NeuroLens: Pre-Flight Creative Diagnostics

NeuroLens addresses this gap with a self-service, pre-flight workflow. Rather than recruiting participants for physical brain scans, it uses Meta’s TRIBE v2 (d’Ascoli et al., 2026), a research model trained to predict average cortical fMRI responses to naturalistic video. The model produces predictions across 20,484 points on the cortical surface; NeuroLens makes this output actionable by summarising it into five region-specific activation signals based on atlas-mapped brain regions (Destrieux et al., 2010):

- Faces (FFA; Kanwisher et al., 1997): Highlights when faces and spokespeople are strongly represented.
- Action & Movement (EBA; Downing et al., 2001): Highlights bodies, gestures, and physical movement.
- Scenes & Settings (PPA; Epstein & Kanwisher, 1998): Highlights locations, environments, and changes of setting.
- Social Interaction (STS; Allison et al., 2000; Frith & Frith, 2006): Highlights gaze, human interaction, and social movement.
- Audio (Binder et al., 2000): Highlights when dialogue, music, and other sounds are strongly represented. A dual-pass silent baseline comparison isolates where the soundtrack changes the cortical response, letting creators review audio timing and compare alternatives.

Commercial licensing: TRIBE v2 is currently released under CC BY-NC 4.0, which does not permit commercial use. Before launching NeuroLens as a commercial service, the project would therefore require either a negotiated commercial licence from Meta or a model whose licence permits commercial deployment. We acknowledge this licensing constraint upfront, but believe the commercial potential and the cloud-architecture challenges it motivates make NeuroLens a worthwhile project for this course regardless of the model’s current licence.

1.4 Target Audience & Competitor Matrix

NeuroLens targets two audiences: independent creators reviewing second-by-second predicted responses to flag underperforming moments for re-editing, and boutique digital agencies comparing creative variants before committing client media spend. Competing platforms such as Neurons and Dragonfly AI predict visual attention from

video features; NeuroLens instead uses a model trained on measured fMRI data, producing multimodal cortical timelines, billed via a credit-pack and subscription model.

Table 1: NeuroLens Competitor Analysis and Cost Comparison

Dimension	Physical fMRI Study	Webcam Facial-Coding	Predictive Analytics	NeuroLens
Underlying Method	Measured fMRI	Computer-vision facial analysis	AI-predicted visual attention	Predicted fMRI responses
Turnaround	Days to weeks	3 hours – 3 days	Seconds to minutes	Seconds to minutes
Indicative Cost	\$10,000+ per study	\$850 – \$4,000 per video	Quote-based pricing	\$0.90 per 30s video*

*Effective cost per video within subscription/credit-pack pricing; not billed per transaction.

2. Business Model & Economics

2.1 The Cloud Economic Justification: Converting CAPEX into OPEX

TRIBE v2 integrates three frozen feature extractors (LLaMA 3.2-3B, V-JEPA 2-Giant, and Wav2Vec-BERT 2.0) with a cortical readout transformer, totalling ~ 5 B parameters and demanding ~ 40 GB of active VRAM during full trimodal inference. This comfortably exceeds commodity 16 GB and 24 GB consumer GPUs. Practical on-premise self-hosting therefore requires an enterprise-grade high-memory workstation:

- The Hardware Penalty: An enterprise workstation equipped with a 48 GB professional GPU (e.g., NVIDIA RTX 6000 Ada) or high-memory accelerator costs approximately \$6,000–\$9,000 upfront, with ongoing power, cooling, and driver maintenance overhead.
- The Utilisation Problem: Ad testing is episodic. An agency may run 20 jobs during an intense editing sprint, then nothing for days. An on-premise GPU sits idle over 80% of the time, inflating the amortised cost per job.
- The Elastic Cloud Remedy: Following Armbrust et al. (2010), NeuroLens pools high-performance cloud GPUs on demand. Queue-depth autoscaling scales workers to zero during off-peak periods, with no idle cost.

2.2 Monetisation Strategy & Unit Economics

We model inference costs on an AWS EC2 `g6e.2xlarge` Spot instance (single 48 GB NVIDIA L40S GPU, $\sim \$0.75/\text{hr}$). On a warm worker, the two-pass analysis of a 30-second ad runs in 28–45 seconds, consuming approximately \$0.008 in raw GPU compute. In practice, cold-start events and idle cooldown periods between jobs raise the blended infrastructure cost to approximately \$0.035 per ad. Factoring in per-request overhead (S3 storage, Aurora writes, ALB traffic) and amortised payment gateway fees, the fully-loaded variable cost is approximately \$0.10 per ad.

To ensure micro-transaction fees do not erode margins, NeuroLens avoids per-analysis billing in favour of a credit-pack and monthly subscription model. Retailing at an average of \$0.90 per ad, the service yields a blended gross margin of approximately 89%. The fixed platform baseline in `ap-southeast-1` (ALB, Aurora Serverless v2, CloudWatch) totals approximately \$70–\$85/month, which is comfortably covered at low subscriber volumes.

Crucially, factoring in the fixed platform baseline, elastic cloud hosting breaks even against a single-GPU on-premise workstation ($\sim \$225/\text{month}$ amortised) at approximately 1,475 videos/month (using the \$0.10/video fully-loaded cost). This suggests that for bursty workflows, a scale-to-zero cloud architecture is vastly more capital-efficient than self-hosting.

2.3 End-to-End User Journey

The five-step workflow is illustrated below: SSO login, direct S3 upload via presigned URL, worker-side duration validation, asynchronous job tracking via polling, and interactive Chart.js visualiser with exportable CSV.

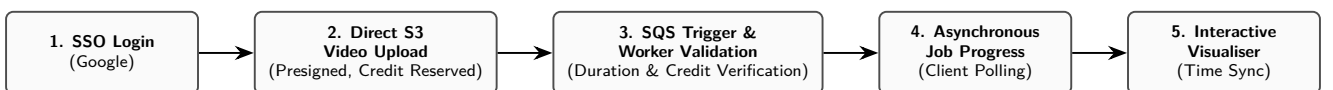


Figure 1: NeuroLens End-to-End User Journey

3. Cloud SaaS Architecture & System Design

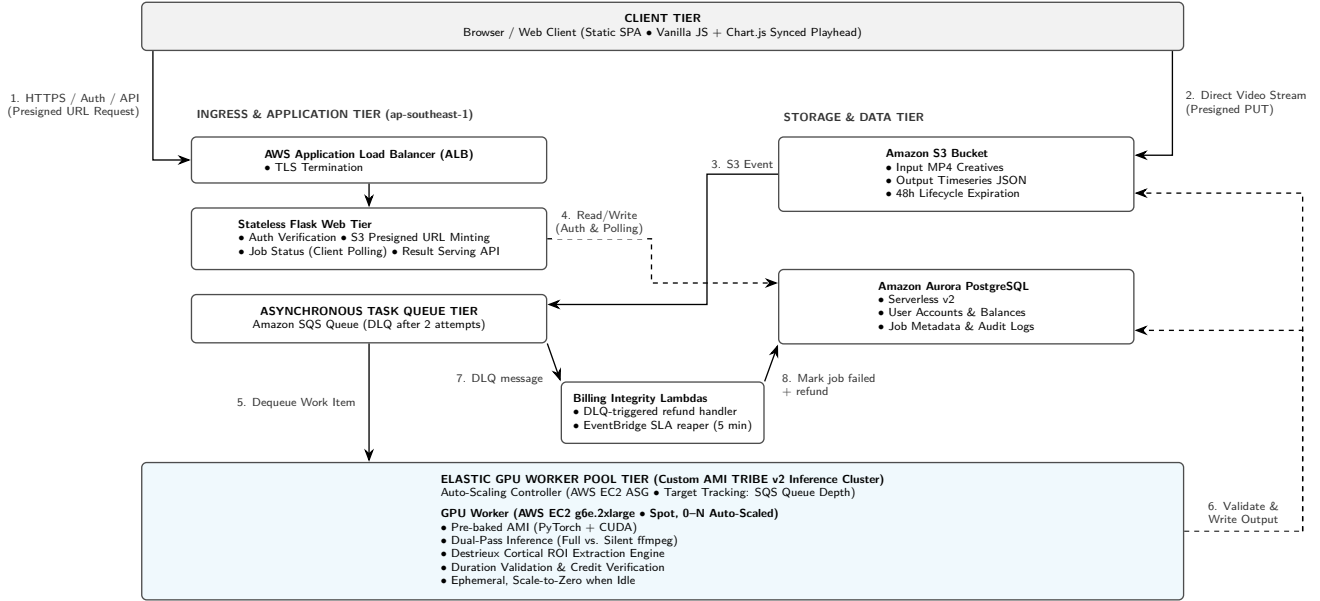


Figure 2: NeuroLens Cloud SaaS Architecture and Data Flow

Every design decision balances the need for elastic scalability with the engineering constraints of an early-stage business. We explicitly prioritised architectural simplicity, strict statelessness, and capital efficiency over premature enterprise optimisation:

- **Ingestion (Separating Data vs. Control Planes):** To separate data and control planes, we routed the heavy video upload stream away from the API web tier: sending 50–250 MB ad creatives through Flask would saturate bandwidth and block concurrent sessions. The web tier instead acts only as an authenticator, minting a presigned S3 URL for direct client upload. To prevent the GPU worker fetching an incomplete file, SQS is triggered natively by the S3 `ObjectCreated` event rather than by Flask. Because this event lacks session context, Flask encodes the User ID into the S3 object key during minting for billing attribution.
- **Compute (Raw EC2 Spot vs. Managed Kubernetes/EKS):** TRIBE v2’s ~40 GB VRAM requirement rules out serverless options like AWS Lambda outright. We weighed raw EC2 Spot against managed Kubernetes (EKS)—which is the enterprise standard but adds steep control-plane costs (~\$73/month) and configuration overhead—and chose an EC2 Auto-Scaling Group on `g6e.2xlarge` Spot instances, cutting compute costs by ~67% versus On-Demand while scaling to zero when idle. The compute tier can later migrate to EKS for a hybrid On-Demand/Spot node group once enterprise SLAs require guaranteed processing times. Spot pricing floats with demand and carries a ~2-minute interruption risk, absorbed via automatic requeueing rather than passed on to the user.
- **Cold-Start Mitigation (Hybrid AMIs vs. Runtime Docker Pulls):** Pulling a 20+ GB container image (PyTorch + CUDA + 5B parameter weights) during an ASG scale-out event introduces 5–8 minutes of latency. To mitigate this, instances boot directly from a custom Amazon Machine Image (AMI) with all heavy dependencies pre-baked. To avoid rebuilding a 30 GB AMI for every minor bug fix, a lightweight `UserData` script pulls the latest Python inference logic on boot. This approach trades the strict immutability of full containerisation for agile code updates and a cold-start latency meaningfully faster than the 5–8 minute container-pull baseline, to be measured empirically in Experiment 2.
- **State Tracking (Stateless Polling vs. WebSockets):** We deliberately kept the web tier strictly stateless. Tracking asynchronous GPU progress is handled via client-side long-polling (every 5 seconds) rather than persistent WebSockets or Server-Sent Events. Despite added DB read volume and minor UI latency, this lets the Application Load Balancer route traffic dynamically to any Flask node without requiring a costly Redis pub/sub broker to manage connection states. As concurrent volume grows and database read-thrashing becomes a concern, the system can migrate to a persistent WebSocket layer backed by Amazon ElastiCache.
- **Validation (Worker Consolidation vs. Microservice Sprawl):** Extracting video duration via `ffprobe` could theoretically run in an S3-triggered Lambda, but compiling `ffprobe` into a Lambda Layer is complex and brittle. We instead moved duration validation into the EC2 GPU worker, avoiding microservice sprawl. This introduces a “denial-of-wallet” vector where low-credit submissions could trigger billable GPU time. Client-side duration estimation and pre-upload credit checks keep most such requests off the GPU tier, with a CPU-only Fargate pre-processing queue as future work once volume demands it.

- **Billing Integrity (Reservation vs. Simple Debit):** A single debit-on-completion model would leave GPU spend uncapped if a user’s balance ran out mid-job. Instead, a reservation is placed at upload, adjusted once the actual duration is verified, and captured on completion, with any unused amount released. Failure recovery is handled automatically by two small, event-driven functions that detect failed or stuck jobs and trigger the corresponding refund, so a lost job never means a silent loss to the user.
- **Payment Security (Hosted Checkout vs. Direct Card Handling):** Handling card data directly introduces meaningful information security risk and compliance burden. We therefore chose Stripe with its standard hosted Checkout flow, which keeps card data off NeuroLens’s own servers entirely and qualifies the platform for the simplest PCI-DSS validation tier (SAQ-A), while still supporting the usage-based, credit-pack billing model described in Section 2.2.
- **Data Tier (ACID Transactions vs. NoSQL):** Financial ledger logic requires strict transactional consistency. DynamoDB’s lower latency comes at the cost of complex `TransactWriteItems` logic for concurrent workflows. We chose Aurora PostgreSQL (Serverless v2) for native ACID guarantees and JSONB indexing, storing job metadata and billing records alongside an S3 pointer to the full result JSON. Serverless v2 auto-pauses at 0 ACUs during idle periods to reduce cost, and can be kept at a 0.5-ACU minimum for responsiveness. An atomic conditional status update ensures at most one worker can claim a queued job, preventing duplicate processing under SQS’s at-least-once delivery. Credentials are managed via AWS Secrets Manager rather than application configuration.

4. Implementation Roadmap

Four milestones deliver the NeuroLens cloud SaaS from initial infrastructure setup to a deployed, evaluated service:

Milestone	Dates	Core Technical Focus & Deliverables
M1	29 Sept – 8 Oct	Cloud Storage & Task Queue Decoupling: • S3 presigned URL generation with size/MIME-type validation; direct client-side upload. • Amazon SQS worker process decoupled from the synchronous Flask handler. • Minimal upload UI: file picker with client-side duration read and estimated-cost display before upload.
M2	9 – 19 Oct	GPU AMI Provisioning & Elastic Autoscaling: • Custom AMI (PyTorch + CUDA 12.4 + TRIBE v2 weights pre-baked) deployed via EC2 Auto Scaling Group, with UserData pulling the latest inference code on boot. • EC2 ASG configured with CloudWatch SQS queue-depth scaling. • Begin latency and Locust load-test data collection (Experiments 1 & 2). • Job progress UI: client-side polling view for queue and processing status.
M3	20 – 30 Oct	Authentication, Billing & Result Persistence: • Google OAuth2 SSO and PostgreSQL schema for user accounts and credit balances. • Credit debiting engine, job history, and CSV result export. • Begin cloud vs. on-premise TCO benchmark runs (Experiment 3). • Google SSO login screen, credit balance/job history dashboard, and Chart.js synced-playhead visualiser.
M4	31 Oct – 13 Nov	Usability Evaluation & Analysis: • 5-participant SUS usability study with task performance metrics. • Consolidation and analysis of all benchmark results for the Final Report.

5. Plan for Evaluation

We plan three technical experiments to validate system performance and cloud economics, and a structured usability study with five participants. While this evaluation plan is designed to thoroughly validate system performance and usability, verifying the predictive fidelity of TRIBE v2’s outputs against real-world ad performance is out of scope for this phase due to data availability, and is noted as future work.

5.1 System Evaluation Plan (Technical & Economic Benchmarks)

1. Experiment 1 — End-to-End Latency Breakdown: Measure per-stage duration (Upload → Queue Wait → Dual Inference → ROI Extraction → Client Render) across 15s, 30s, and 60s video lengths. Inference duration and peak VRAM are logged automatically by the pipeline.
2. Experiment 2 — Elastic Scaling & Throughput Under Load: Using Locust, simulate concurrent submission bursts (1, 5, 10, 20 jobs) in two stages: lower bursts (1, 5) measure GPU cold-start latency and per-worker throughput as the worker pool scales with demand; higher bursts (10, 20) confirm the queue absorbs backlog gracefully once autoscaling capacity is saturated, with a full drain once load subsides. A Spot interruption will be manually injected to validate the SQS requeue mechanism.
3. Experiment 3 — Cloud vs. On-Premise TCO Benchmark: Execute 10–15 benchmark runs per video duration (15s, 30s, 60s) on cloud Spot instances and available team hardware (~60–90 runs total). Power consumption is modelled from published hardware thermal design power (TDP) and measured inference durations. Results use measured inference runtimes to estimate cloud and on-premise costs, then compare those estimates with the Section 2.2 breakeven point.

5.2 User Evaluation Plan (SaaS Usability & Qualitative Feedback)

- Participant Cohort (N=5): Following Nielsen’s finding that five participants uncover approximately 85% of usability issues in formative testing (Nielsen, 2000), we recruit 5 users from the target demographic — student content creators and peers managing digital media channels. Participants will provide informed consent covering upload of self-selected ad samples.
- End-to-End Creative Audit Scenario: Participants complete one end-to-end workflow: uploading an ad creative, observing real-time cloud processing progress, and exploring the synchronised playhead timeline to identify attention peaks and drop-off moments.
- Evaluation Metrics & Qualitative Feedback:
 - System Usability Scale (SUS; Brooke, 1996): Standardised 10-item questionnaire targeting the Grade A threshold ($\geq 80/100$); scores below 80 will inform targeted future UI refinements.
 - Task Performance: Task success rate and time-on-task for the core creative audit workflow.
 - Structured Qualitative Feedback: Open-ended evaluation capturing (1) dashboard clarity and playhead responsiveness, (2) comprehension of cortical ROI labels, and (3) perceived commercial value of the \$0.90/ad pricing model.

6. Conclusion

NeuroLens targets a market that current tools underserve, delivering cloud-native cortical video analysis as a credit-pack and subscription-based service. Every architectural choice is driven by a concrete constraint: Spot GPU workers for elastic compute economics, SQS queue decoupling for fault-tolerant async processing, Aurora PostgreSQL for ACID credit integrity, and presigned S3 ingestion to keep the web tier stateless.

The unit economics in Section 2.2 demonstrate that elastic cloud hosting remains cost-effective below approximately 1,475 videos/month, eliminating prohibitive hardware CAPEX for target users while scaling efficiently to zero during idle periods. Overall, the architecture provides a robust, fault-tolerant foundation for scalable video engagement prediction, with empirical performance and usability evaluations scheduled across the upcoming milestone phases.

AI Declaration: AI tools were used to assist in refining the prose, as a thought partner to evaluate the cloud architectural trade-offs in Section 3, and to help structure the evaluation plan in Section 5. This process deepened the team’s practical understanding of both cloud-native system design and robust testing methodologies. However, all final technical design decisions, cost analyses, and evaluation methodologies were ultimately determined and validated by the project team.

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